

Math Foundations: Probability and Statistics

Paired reading: Wasserman, *All of Statistics*, Part 1

Errata: 02_readings/00_prerequisites/errata2.pdf **Focus area A** (flagged weak area)

This underlies OL4–OL5 (Bayesian learning), UL1’s EM, and, crucially, the experimental analysis in every single report. You cannot claim a result is “real” without the vocabulary here. Log gaps in `math_gaps_log.md`.

1 Probability Axioms: The Floor Everything Is Built On

A **probability space** is $(\Omega, \mathcal{F}, \mathbb{P})$:

- Ω : sample space, the set of outcomes.
- $\mathcal{F}$: set of events, subsets of Ω .
- $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$: probability measure satisfying **Kolmogorov’s axioms**.

$$\begin{aligned} \text{(A1)} \quad & \mathbb{P}(A) \geq 0 && \text{for every event } A, \\ \text{(A2)} \quad & \mathbb{P}(\Omega) = 1 && \text{something happens,} \\ \text{(A3)} \quad & \mathbb{P}(A_1 \cup A_2 \cup \dots) = \sum_i \mathbb{P}(A_i) && \text{for disjoint events.} \end{aligned}$$

Everything else, including Bayes, expectation, MLE, and the bias–variance decomposition, is algebra on top of these three rules. Two useful consequences:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \quad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

2 Random Variables and Distributions

A **random variable** is a function $X : \Omega \rightarrow \mathbb{R}$ that assigns a number to each outcome. Its **distribution** is how probability mass or density is spread:

- **Discrete:** probability mass function $p_X(x) = \mathbb{P}(X = x)$, with $\sum_x p_X(x) = 1$.
- **Continuous:** probability density function $p_X(x) \geq 0$, with $\int p_X(x) dx = 1$. Note: $p_X(x)$ is not a probability; it is a density. $\mathbb{P}(a \leq X \leq b) = \int_a^b p_X(x) dx$.

The **cumulative distribution function (CDF)** is $F(x) = \mathbb{P}(X \leq x)$, defined for both. F is right-continuous and non-decreasing, with $F(-\infty) = 0$ and $F(\infty) = 1$.

2.1 Distributions to Recognize on Sight

Distribution	Parameters	PMF / PDF	Mean	Var
Bernoulli(p)	$p \in [0, 1]$	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	n, p	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Categorical(π)	$\pi \in \Delta^{K-1}$	$\prod_k \pi_k^{\mathbf{1}[x=k]}$	–	–
Gaussian $\mathcal{N}(\mu, \sigma^2)$	μ, σ^2	$\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	μ	σ^2
Uniform($[a, b]$)	$a < b$	$\frac{1}{b-a}$ on $[a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Beta(α, β)	$\alpha, \beta > 0$	$\propto x^{\alpha-1}(1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
Exponential(λ)	$\lambda > 0$	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Multivariate Gaussian	$\mu \in \mathbb{R}^d, \Sigma \text{ PSD}$	$(2\pi)^{-d/2} \det(\Sigma)^{-1/2} \times \exp\left(-\frac{1}{2}\delta^\top \Sigma^{-1} \delta\right), \delta = x - \mu$	μ	Σ

Why these matter for CS7641:

- **Gaussian:** the noise model implicit in least-squares (OL4) and the per-component model in GMM/EM (UL1).
- **Bernoulli / Categorical:** the output distribution for binary/multiclass classification; logistic regression and softmax are literally parametrizing these.
- **Beta:** the conjugate prior for Bernoulli; OL4 MAP examples often live here.
- **Multivariate Gaussian:** what PCA assumes implicitly; what GMM's components are.

3 Expectation, Variance, Covariance

3.1 Expectation

$$\mathbb{E}[X] = \sum_x x p_X(x) \quad (\text{discrete}), \quad \mathbb{E}[X] = \int x p_X(x) dx \quad (\text{continuous}).$$

The single most-used identity in the course: **linearity of expectation**

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c \quad (\text{always, dependent or not}).$$

For any function g , the **law of the unconscious statistician**:

$$\mathbb{E}[g(X)] = \sum_x g(x) p_X(x),$$

with the analogous integral form for continuous X . No need to derive $g(X)$'s distribution.

3.2 Variance

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Derivation of the right form, which you should be able to reproduce in your sleep:

$$\begin{aligned} \mathbb{E}[(X - \mu)^2] &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \mathbb{E}[X^2] - 2\mu^2 + \mu^2 \\ &= \mathbb{E}[X^2] - \mu^2. \end{aligned}$$

Properties:

$$\text{Var}(aX + b) = a^2 \text{Var}(X), \quad \text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) \quad \text{iff } X \text{ and } Y \text{ are uncorrelated; independence suffices.}$$

3.3 Covariance and Correlation

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y],$$

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

The **covariance matrix** of a random vector $X = (X_1, \dots, X_d)^\top$:

$$\Sigma_{ij} = \text{Cov}(X_i, X_j), \quad \Sigma = \mathbb{E}[(X - \mu)(X - \mu)^\top].$$

Σ is symmetric, PSD, and is the object PCA eigen-decomposes.

3.4 Independence vs. Uncorrelated: The Distinction That Matters

- **Independent:** $p_{X,Y}(x, y) = p_X(x)p_Y(y)$, so the joint factorizes.
- **Uncorrelated:** $\text{Cov}(X, Y) = 0$, so there is no linear relationship.

Independent implies uncorrelated, but the converse fails. Canonical counterexample: $X \sim \text{Uniform}[-1, 1]$, $Y = X^2$. Then

$$\text{Cov}(X, Y) = \mathbb{E}[X \cdot X^2] - \mathbb{E}[X]\mathbb{E}[X^2] = \mathbb{E}[X^3] - 0 = 0,$$

but they are maximally dependent because Y is a function of X .

Where this hits CS7641:

- **PCA vs. ICA (UL3):** PCA only de-correlates, meaning zero off-diagonal covariance; ICA seeks true independence. The example above is precisely the regime where PCA fails and ICA wins.
- **Naive Bayes (OL5):** the “naive” conditional independence assumption is strong and routinely violated, but the model still works because the decision rule only needs relative posteriors, not calibrated probabilities.

4 Conditional Probability and Bayes’ Rule

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A, B)}{\mathbb{P}(B)} \quad \text{requires } \mathbb{P}(B) > 0,$$

$$\mathbb{P}(A, B) = \mathbb{P}(A \mid B)\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A).$$

Bayes’ rule rearranges:

$$\mathbb{P}(H \mid D) = \frac{\mathbb{P}(D \mid H)\mathbb{P}(H)}{\mathbb{P}(D)}.$$

In learning vocabulary:

$$\underbrace{\mathbb{P}(H | D)}_{\text{posterior}} = \frac{\overbrace{\mathbb{P}(D | H)}^{\text{likelihood}} \overbrace{\mathbb{P}(H)}^{\text{prior}}}{\underbrace{\mathbb{P}(D)}_{\text{evidence}}}.$$

The **evidence** $\mathbb{P}(D)$ is just a normalizer, independent of H , computed via the **law of total probability**:

$$\mathbb{P}(D) = \sum_h \mathbb{P}(D | h) \mathbb{P}(h),$$

or an integral for continuous H . When comparing two hypotheses h_1, h_2 , $\mathbb{P}(D)$ cancels, which is why MAP estimation never needs to compute it explicitly.

4.1 Worked Bayes Example: The Canonical Disease/Test Problem

A disease has prevalence $\mathbb{P}(D) = 0.001$. A test has sensitivity $\mathbb{P}(+ | D) = 0.99$ and specificity $\mathbb{P}(- | \neg D) = 0.95$. Given a positive test, what is $\mathbb{P}(D | +)$?

$$\begin{aligned} \mathbb{P}(+ | \neg D) &= 1 - 0.95 = 0.05, \\ \mathbb{P}(+) &= \mathbb{P}(+ | D) \mathbb{P}(D) + \mathbb{P}(+ | \neg D) \mathbb{P}(\neg D) \\ &= 0.99 \cdot 0.001 + 0.05 \cdot 0.999 \\ &= 0.00099 + 0.04995 \\ &= 0.05094, \\ \mathbb{P}(D | +) &= \frac{\mathbb{P}(+ | D) \mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.00099}{0.05094} \approx 0.0194. \end{aligned}$$

The posterior is about 2%, not 99%. The intuition: a 5% false-positive rate applied to 99.9% of the population generates 50 times more false positives than the rare true positives. **Base rates dominate**. This is why “test accuracy” is the wrong metric for rare conditions, a critical thread to pull through every classification report.

4.2 Bayesian Networks (OL5 Preview)

A Bayes net is a DAG whose nodes are random variables and edges encode conditional dependence. The joint factorizes as

$$\mathbb{P}(X_1, \dots, X_n) = \prod_i \mathbb{P}(X_i | \text{parents}(X_i)).$$

Independence claims, read off the graph via d-separation, collapse joints into products of conditionals. That is the whole exam-relevant payoff.

5 MLE, MAP, and the Bayesian View

Two ways, three really, to estimate parameters θ from data $D = \{x_1, \dots, x_n\}$:

5.1 Maximum Likelihood (MLE)

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} \mathbb{P}(D \mid \theta) = \arg \max_{\theta} \log \mathbb{P}(D \mid \theta) = \arg \max_{\theta} \sum_i \log \mathbb{P}(x_i \mid \theta),$$

where the final equality assumes i.i.d. data.

We work with **log-likelihood** because products turn into sums, making derivatives tractable, and because tiny probabilities multiplied a million times underflow to zero in float64.

Worked MLE: Bernoulli. Data $D = (x_1, \dots, x_n)$ with $x_i \in \{0, 1\}$ and parameter p . Log-likelihood:

$$\ell(p) = \sum_i [x_i \log p + (1 - x_i) \log(1 - p)] = k \log p + (n - k) \log(1 - p), \quad k = \sum_i x_i.$$

Set $d\ell/dp = 0$:

$$\frac{k}{p} - \frac{n - k}{1 - p} = 0 \quad \Rightarrow \quad k(1 - p) = (n - k)p \quad \Rightarrow \quad k = np \quad \Rightarrow \quad \hat{p}_{\text{MLE}} = \frac{k}{n}.$$

Sanity: this is exactly what intuition says; the MLE of a coin's bias is the observed frequency.

Worked MLE: Gaussian mean with known σ^2 .

$$\ell(\mu) = \sum_i \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(x_i - \mu)^2}{2\sigma^2} \right],$$

$$\frac{d\ell}{d\mu} = \frac{1}{\sigma^2} \sum_i (x_i - \mu) = 0 \quad \Rightarrow \quad \hat{\mu}_{\text{MLE}} = \frac{1}{n} \sum_i x_i = \bar{x}.$$

The MLE of the mean is the sample mean. Equivalently, minimizing $\sum_i (x_i - \mu)^2$ is exactly MLE under a Gaussian noise assumption. That equivalence, least-squares equals Gaussian MLE, is the OL4 load-bearing fact.

5.2 Maximum A Posteriori (MAP)

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \mathbb{P}(\theta \mid D) = \arg \max_{\theta} \mathbb{P}(D \mid \theta) \mathbb{P}(\theta) = \arg \max_{\theta} [\log \mathbb{P}(D \mid \theta) + \log \mathbb{P}(\theta)].$$

MLE = MAP with a uniform prior: $\log \mathbb{P}(\theta)$ is constant and drops out.

Why this matters: many regularizers are MAP priors in disguise:

Regularizer	Equivalent prior
L_2 (Ridge): $\lambda \ \beta\ ^2$	Gaussian $\beta \sim \mathcal{N}(0, \sigma^2 I)$
L_1 (Lasso): $\lambda \ \beta\ _1$	Laplace $\beta \sim \text{Laplace}(0, b)$
$L_2 + L_1$ (Elastic net)	Mixture of Gaussian and Laplace

So **regularized regression is MAP estimation**. This collapses two seemingly different ideas, penalty terms and priors, into one, and gives you a principled way to interpret regularization strength.

5.3 Full Bayesian Inference

MLE and MAP give point estimates. Full Bayesian inference keeps the entire posterior:

$$\mathbb{P}(\theta \mid D) = \frac{\mathbb{P}(D \mid \theta)\mathbb{P}(\theta)}{\mathbb{P}(D)},$$

$$\mathbb{P}(y^* \mid x^*, D) = \int \mathbb{P}(y^* \mid x^*, \theta)\mathbb{P}(\theta \mid D) d\theta \quad (\text{posterior predictive}).$$

Usually intractable, which is why EM, variational inference, and MCMC exist.

6 EM (UL1 Preview)

When you have **latent variables** z , for example “which Gaussian did this point come from?”, the likelihood $\sum_z \mathbb{P}(x, z \mid \theta)$ becomes a sum inside the log, killing the closed form. EM alternates:

- **E-step:** compute $q(z) = \mathbb{P}(z \mid x, \theta_{\text{old}})$, a soft assignment.
- **M-step:** maximize $\mathbb{E}_q[\log \mathbb{P}(x, z \mid \theta)]$ with respect to θ , a weighted MLE that usually has a closed form.

Each iteration provably increases the marginal log-likelihood, but only to a local optimum.
k-means is the hard-assignment limit of EM on a GMM with isotropic covariances.

7 Sampling, Estimators, and the Report-Critical Statistics

Your training set is one **sample** from a population. Any statistic computed on it, such as accuracy, mean, or model score, is an **estimate** with its own uncertainty.

7.1 Bias, Variance, MSE of an Estimator

For an estimator $\hat{\theta}$ of a true parameter θ :

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta, \quad \text{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2],$$

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta}).$$

The bias–variance decomposition for an estimator is structurally identical to SL1’s bias–variance decomposition for a model: same algebra, different object.

7.2 Standard Error

If $X_1, \dots, X_n$ are i.i.d. with variance σ^2 , then $\text{Var}(\bar{X}) = \sigma^2/n$, so

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}},$$

or $s/\sqrt{n}$ in practice, using the sample standard deviation.

Standard error shrinks as $1/\sqrt{n}$: diminishing returns to data. Doubling your dataset only multiplies precision by $\sqrt{2} \approx 1.41$. To halve your standard error you need **4 times the data**. This is exactly the shape of a learning curve in SL1: a steep initial drop, then a long flat tail.

7.3 Central Limit Theorem

For i.i.d. $X_1, \dots, X_n$ with finite mean μ and variance σ^2 :

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \rightarrow \mathcal{N}(0, 1) \quad \text{as } n \rightarrow \infty.$$

This is why we can approximate distributions of means by Gaussians for large enough n , and the basis of every confidence interval and t-test you will construct.

7.4 Confidence Intervals

A 95% CI for the mean, using large n and a normal approximation:

$$\bar{X} \pm 1.96 \cdot \frac{s}{\sqrt{n}}.$$

Correct interpretation: if we repeated this experiment many times, constructing a CI each time by this procedure, about 95% of the intervals would contain the true μ . Not “there is a 95% chance the true value is in this interval”; that is a Bayesian credible interval, a different object.

For small n , or unknown σ , use the **t-distribution** with $n - 1$ degrees of freedom: replace 1.96 with $t_{\alpha/2, n-1}$.

7.5 Hypothesis Testing and p-Values

The setup:

- H_0 (null hypothesis): often “no effect” or “no difference.”
- H_1 (alternative): the thing you suspect.
- **Test statistic** T : a function of the data whose distribution under H_0 is known.
- **p-value:** $\mathbb{P}(T \geq T_{\text{obs}} \mid H_0)$, the probability of seeing data this extreme if H_0 were true.

$$\text{p-value} = \mathbb{P}(\text{data this extreme} \mid H_0 \text{ true}).$$

The p-value is **not** $\mathbb{P}(H_0 \mid \text{data})$. The classic misreading “ $p = 0.03$ means there is a 97% chance our model is better” inverts the conditional. To actually compute $\mathbb{P}(H_0 \mid \text{data})$ you would need a prior and Bayes’ rule.

7.6 Comparing Two Classifiers: What to Actually Report

For the report-grading rubric, “Model A scored 0.84, Model B 0.83” is meaningless without uncertainty. Use one of:

Paired t-test on cross-validation folds, if you ran both models on the same k folds:

$$d_i = \text{score}_{A,i} - \text{score}_{B,i}, \quad t = \frac{\bar{d}}{s_d/\sqrt{k}} \sim t_{k-1}.$$

Caveat: CV folds are not fully independent because samples overlap across folds, so the test is slightly anti-conservative. Use with $k \geq 10$ and report the effect size $\bar{d}$, not just significance.

McNemar’s test, for two classifiers on the same held-out test set:

	B correct	B wrong
A correct	a	b
A wrong	c	d

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \sim \chi^2(1).$$

Tests whether the two classifiers disagree symmetrically. The diagonal (a, d) does not enter; only disagreements matter.

5x2 CV paired t-test (Dietterich) is the gold standard when computationally feasible, designed specifically to control type-I error in algorithm comparisons.

For your reports, **always** report:

- Mean $\pm$ standard deviation, or SE, across at least 5 seeded runs.
- Confidence intervals or error bars in plots.
- A significance test when ranking models, and the effect size, not just p .

8 Where It Shows Up in CS7641

Concept	Where
Bayes’ rule, priors/posteriors	OL4, OL5, UL1 (EM)
MLE / MAP, log-likelihood	OL4 (least-squares, regularization)
MLE equals least-squares under Gaussian noise	OL4 squared error
L_2/L_1 regularization as MAP	OL4 ridge/lasso
Gaussian noise model	OL4; UL1 GMM components
Independence vs. uncorrelated	OL5 Naive Bayes; UL3 PCA vs. ICA
Variance additivity (independence)	SL5 bagging; averaging seeded runs
Bias–variance decomposition	SL1
CLT, CIs, paired t-test, McNemar	Every report’s experimental analysis

9 Review Questions with Sketched Answers

Conceptual Checks

1. Independent vs. uncorrelated, with a concrete example.

Independent means $p(x, y) = p(x)p(y)$, so the joint factorizes. Uncorrelated means $\text{Cov}(X, Y) = 0$, no linear relationship. Counterexample: $X \sim U[-1, 1]$, $Y = X^2$. Then $\text{Cov} = \mathbb{E}[X^3] = 0$, but Y is a deterministic function of X . This hinges PCA vs. ICA (UL3).

2. Bayes’ rule, with all four terms and the normalizer.

$\mathbb{P}(H \mid D) = \mathbb{P}(D \mid H)\mathbb{P}(H)/\mathbb{P}(D)$. Likelihood times prior over evidence equals posterior. $\mathbb{P}(D)$ is the evidence/normalizer; it does not depend on H , so it drops out when comparing hypotheses (MAP).

3. Precise relationship between MLE and MAP.

MLE = MAP with a uniform prior: when $\mathbb{P}(\theta)$ is constant, $\log \mathbb{P}(\theta)$ contributes nothing to the argmax, so $\arg \max \mathbb{P}(D \mid \theta)\mathbb{P}(\theta) = \arg \max \mathbb{P}(D \mid \theta)$.

Derivation / Math

4. **Derive** $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 = \mathbb{E}[X^2] - \mu^2$. This uses linearity of expectation.

5. **Why** $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ **only for independent variables, while** $\mathbb{E}[X + Y]$ **is always additive.**

In general, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$. Expectation is linear, a stronger property, so cross terms cancel regardless. Variance is quadratic: cross terms vanish only under independence, or at least uncorrelatedness. This is why bagging, averaging n independent models, reduces variance by $1/n$: independence makes the variances add, then the $1/n^2$ from averaging does the rest.

6. **Two reasons to maximize log-likelihood.**

Products become sums, so derivatives decompose cleanly, including gradients in SGD. Log-likelihood is also numerically stable: $\prod_i p_i$ underflows, while $\sum_i \log p_i$ does not. Since log is monotonic, the argmax is unchanged.

Analysis: Why Does It Matter?

7. **SE proportional to $1/\sqrt{n}$: implication for 4x data and learning curves.**

Four times the data gives a two times tighter SE. Early data points buy a lot of precision; the marginal gain falls quickly. Learning curves show the same shape: fast early improvement, then a long tail of diminishing returns.

8. **“ $p = 0.03$, so 97% chance our model is better.”**

This inverts the conditional. $p = \mathbb{P}(\text{data} \mid H_0)$, not $\mathbb{P}(H_0 \mid \text{data})$. The correct reading: under the null hypothesis of no real difference, we would see results this extreme 3% of the time. To get a probability over hypotheses you would need a prior and Bayes’ rule.

9. **Why single-run reporting is indefensible for stochastic algorithms.**

The result you saw is a single sample from a distribution over outcomes induced by initialization, random splits, and similar sources. It could be a lucky run. **Always** report mean $\pm$ SD or SE across multiple seeded runs, preferably with a CI and a significance test against the baseline.

Exam-Style

10. **Show** $h_{\text{MAP}} = h_{\text{MLE}}$ **under a uniform prior.**

$h_{\text{MAP}} = \arg \max_h \mathbb{P}(D \mid h)\mathbb{P}(h)$. If $\mathbb{P}(h) = c$ constant, this equals $\arg \max_h c\mathbb{P}(D \mid h) = h_{\text{MLE}}$. The uniform prior represents no prior information: every hypothesis is equally likely a priori.

11. **Bias/variance of an estimator vs. of a model.**

Same algebraic decomposition: $\text{MSE} = \text{Bias}^2 + \text{Var}$. For an estimator, bias/variance describe how $\hat{\theta}$ jumps around across samples relative to the true θ . For a model (SL1), they describe how predictions jump around across training sets relative to the true target function. Identical math, one object up.

12. **Procedure for comparing two classifiers in a report.**

Run both models with at least 5 seeds times k -fold CV, or 5x2 CV. Report per-fold accuracies, compute paired differences, and run a paired t-test, or McNemar if on a fixed test set. Put **mean** $\pm$ **SE** in the table and **error bars** on every plotted point. State the effect size, p-value, and the test used.